

Integrating Indigenous Knowledge and Socio-Ecological Adaptation for Community Resilience to Compound Climate Hazards in Khagrachhari, Bangladesh

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Abstract:

Khagrachhari district in the Chittagong Hill Tracts is recognized as environmental vulnerable hotspot, increasingly exposed to several compound climatic hazards, including landslides, flash floods, droughts, forest degradation, water scarcity etc. Existing studies highly focus on hazard-specific approaches, forest deforestation, often overlook Indigenous knowledge that have sustained socio-ecological, community resilience and adaption for generations. This study examines how Indigenous knowledge and socio-ecological adaptation routes contribute to community resilience against multiple interacting hazards in Khagrachhari. This study integrates secondary climate hazards data, peer-reviewed literature. The findings are expected to demonstrate that Indigenous adaptation practices, including community forest and traditional resource management, enhance resilience by strengthening ecosystem services and reducing vulnerability risks. The study provides a novel foundation for integrating Indigenous knowledge into local climatic adaptation and disaster risk reduction (DRR) strategies in Bangladesh.

Keywords: *Chittagong Hill Tracts, compound climatic hazards, forest degradation, community resilience, Indigenous knowledge, socio-ecological adaptation, Indigenous adaptation practices, disaster risk reduction (DRR).*